INFO20003 Database Systems

Lecturer: Egemen Tanin

Lecture 16

Week 8

Armstrong's Axioms

Functional dependencies can be identified using Armstrong's Axioms

$A = (X1, X2, \ldots, Xn)\ and\ B = (Y1, Y2, \ldots, Yn)$

1. Reflexivity: $B \subseteq A \Longrightarrow A \rightarrow B$

   Example: Student_ID, name -> name

2. Augmentation: $A \rightarrow B \Longrightarrow \mathrm{AC} \rightarrow BC$

   Example: Student_ID -> name => Student_ID, surname ->name, surname

3. Transitivity: $A \rightarrow B \text{ and } \mathrm{B} \rightarrow C \Longrightarrow A \rightarrow C$

   Example: ID -> birthdate and birthdate -> age then ID ->age

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Steps in Normalisation

First Normal Form	Keep atomic data/Remove repeating groups
↓	
Second Normal Form	Remove partial dependencies
↓	
Third Normal Form	Remove transitive dependencies

… and there are even further NFs but not all are used!

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First Normal Form

Remove Repeating Groups

- repeating groups of attributes cannot be represented in a flat, two dimensional table
- removing cells with multiple values (keep atomic data)

Set of values

Example: Order-Item (Order#, Customer#, (Item#, Desc, Qty))

- Order-Item (Order#, Customer#, (Item#, Desc, Qty))

↓

- Order-Item (Order#, Item#, Desc, Qty)
- Order (Order#, Customer#)

Break them into two
Use PK/FK to connect

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Second Normal Form

Remove Partial Dependencies

a non-key attribute cannot be identified

by part of a composite key

Example: Order-Item (Order#, Item#, Desc, Qty)

- Order-Item (Order#, Item#, Desc, Qty)

Partial dependency

Item (Item#, Desc)

Order-Item (Order#, Item#, Qty)

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Partial Dependency Anomalies

Order-Item (Order#, Item#, Desc, Qty)

Order#	Item#	Desc	Qty
27	873	nut	2
28	402	bolt	1
28	873	nut	10
30	495	washer	50

- *UPDATE* change item desc in many places
- *DELETE* data for last item lost when last order for that item is deleted
- *INSERT* cannot add new item until it is ordered

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Solution to these Anomalies

Order-Item

Order#	Item#	Qty
27	873	2
28	402	1
28	873	10
30	495	50

delete last order for item, but item remains

add new item at any time

Item

Item#	Desc
873	nut
402	bolt
495	washer

change item description in one place

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Third Normal Form

Remove Transitive Dependencies

a non-key attribute **cannot be identified**

by another non-key attribute

Example: Employee (Emp#, Ename, Dept#, Dname)

- Employee (Emp#, Ename, Dept#, Dname)

Transitive dependency

- Employee (Emp#, Ename, Dept#)
- Department(Dept#, Dname)

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Transitive Dependency Anomalies

Example: Employee (Emp#, Ename, Dept#, Dname)

Emp#	Ename	Dept#	Dname
10	Smith	D5	MIS
20	Jones	D7	Finance
25	Smith	D7	Finance
30	Black	D8	Sales

- *UPDATE* change dept name in many places
- *DELETE* data for dept lost when last employee for that dept is deleted
- *INSERT* cannot add new dept until an employee is allocated to it

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Solution to these Anomalies

Employee

Emp#	Ename	Dept#
10	Smith	D5
20	Jones	D7
25	Smith	D7
30	Black	D8

delete last emp in dept, but dept remains

add new dept at any time

Dept#	Dname
D5	MIS
D7	Finance
D8	Sales

change dept name in one place

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Boyce-Codd NF and Anomalies

Every determinant must be a candidate key

Allocation

Stud#	Subject	Teacher
123	Physics	Einstein
123	Music	Mozart
456	Biology	Darwin
789	Physics	Bohr
999	Physics	Einstein

- *UPDATE* student changing subject may lose teacher for that subject
- *DELETE* deleting student may lose teacher for that subject
- *CREATE* cannot assign a teacher to a subject until a student takes it

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Solution to these Anomolies

Stud#	Teacher
123	Einstein
123	Mozart
456	Darwin
789	Bohr
999	Einstein

delete student but subject remains

Teacher	Subject
Einstein	Physics
Mozart	Music
Darwin	Biology
Bohr	Physics

change enrolment without losing teacher/subject

assign teacher to subject

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NFs are not a magic bullet!

- Normalization does not fix all design flaws. For example…
- Design 1:
  - Earnings_2007(company,amount),
  - Earnings_2008(company,amount),
  - Earnings_2009(company,amount)…

  - Tables are in BCNF, but design is cumbersome, redundant, some queries will be difficult.

- Design 2:
  - Earnings(company,year,amount)

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Normalisation vs Denormalization

Normalisation:

- Normalised relations contains a minimum amount of redundancy and allow users to insert, modify, and delete rows in tables without errors or inconsistencies (anomalies)

Denormalization:

- The pay-off: query speed.
- The price: extra work on updates to keep redundant data consistent.
- Denormalization may be used to improve performance of time-critical operations.

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Homework

Report (reportNo, editor, deptNo, deptName, deptAddr, authorId, authName, authAddr)

reportNo → editor, deptNo

deptNo → deptName, deptAddr

authorId → authName, authAddr

reportNo	editor	deptNo	deptName	deptAddr	authorId	authName	authAddr
4216	woolf	15	design	argus1	53	mantel	cs-tor
4216	woolf	15	design	argus1	44	bolton	mathrev
4216	woolf	15	design	argus1	71	koenig	mathrev
5789	koenig	27	analysis	argus2	26	fry	folkstone
5789	koenig	27	analysis	argus2	38	umar	prise
5789	koenig	27	analysis	argus2	71	koenig	mathrev

- Is the Report table in 2NF? If not, put the table in 2NF.
- Are there any insert, update or delete anomalies with these 2NF relations?

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More Homework

Class (courseNumber, roomNumber, instructorName, studentNumber, workshopNumber, grade, tutor)

workshopNumber → tutor

studentNumber, courseNumber → grade, workshopNumber

courseNumber → roomNumber, instructorName

- Normalise the relation into 3NF.

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